INCH-POUND

MIL-DTL-83725/9C
01 October 2018

SUPERSEDING
MIL-R-83725/9B(USAF)
26 November 1986

DETAIL SPECIFICATION

# RELAYS, VACUUM, SPST, LATCHING, 12 AMPERES DC OR 60 Hz RMS, 7 KILOVOLTS PEAK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-DTL-83725.

Inactive for new design after 01 October 2018.

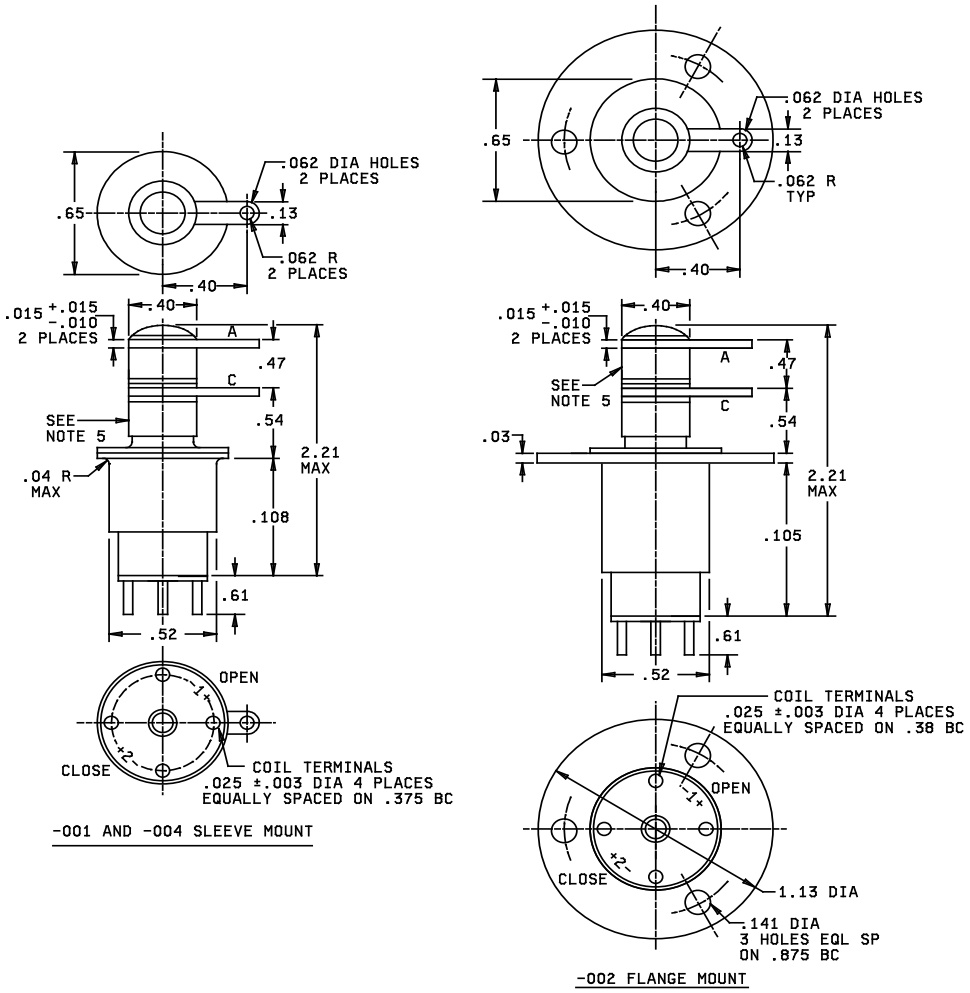
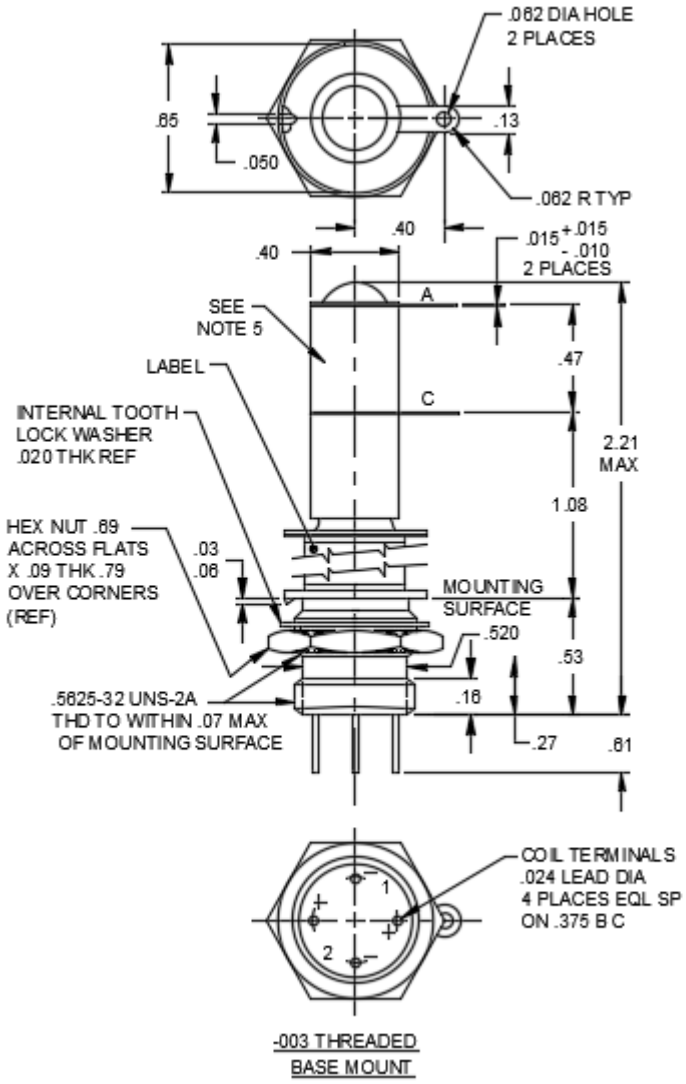


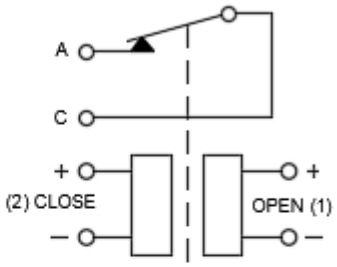

FIGURE 1. Relay, dimensions and configurations.

AMSC N/A

FSC 5945

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CIRCUIT DIAGRAM

| Inches | mm | Inches | mm |
|---|---|---|---|
| .003 | 0.08 | 0.38 | 9.65 |
| .015 | 0.38 | 0.40 | 10.2 |
| .020 | 0.51 | 0.47 | 11.9 |
| .024 | 0.61 | 0.52 | 13.2 |
| .025 | 0.64 | 0.53 | 13.5 |
| .03 | 0.76 | 0.54 | 13.7 |
| .04 | 1.02 | .5625 | 14.29 |
| .05 | 1.27 | .57 | 14.5 |
| .06 | 1.52 | .61 | 15.5 |
| .062 | 1.57 | .65 | 16.5 |
| .07 | 1.78 | .69 | 17.5 |
| .09 | 2.29 | .79 | 20.0 |
| .13 | 3.30 | .875 | 22.2 |
| .141 | 3.58 | 1.05 | 26.7 |
| .16 | 4.06 | 1.08 | 27.4 |
| .27 | 6.86 | 1.13 | 28.7 |
| .375 | 9.53 | 2.21 | 56.1 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three-place decimals and ±.03 (0.8 mm) for two-place decimals.
4. Shape of coil lug (solder) terminals optional; however, they must accommodate two no. 22 AWG wires.
5. Glazed finish on ceramic insulators is optional.
6. M83725/9-001 and -004 mounted by 0.52 inch (13.2 mm) diameter of body of coil housing or by 0.65 inch (16.5 mm) diameter lip in center of relay.

FIGURE 1. Relay, dimensions and configurations - Continued.



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REQUIREMENTS:

Contact data:

Configuration: SPST, latching, ground isolated.

Load ratings:

Resistive (carry only):
12 amperes dc or 60 Hz rms.
10 amperes rms, 2.5 MHz.
6 amperes rms, 16 MHz (-001, -002, and -003).
8 amperes rms, 16 MHz (-004).

Rated operating voltage:
7 kilovolts peak, 60 Hz or dc.
7 kilovolts peak, 2.5 MHz (-001, -002, and -003).
6 kilovolts peak, 16 MHz.
9 kilovolts peak, 2.5 MHz (-004).

Contact resistance:

Rated Life:

Before: .020 ohm, maximum.

During: .100 ohm, maximum.

After: .100 ohm, maximum.

Capacitance:
1.6 picofarads, maximum across open contacts.
1.6 picofarads, maximum between open contacts and ground.

Coil data:

Duty rating: Intermittent duty.

Maximum voltage: 32 V dc.

Nominal voltage: 26.5 V dc. Minimum pulse length, 5 ms; maximum pulse length, 50 ms.

Transfer voltage:
22 V dc, maximum over temperature range (-001, -002, and -003).
16 V dc, maximum at 25°C.
19 V dc, maximum over 40°C to 71°C temperature range (-004).

Coil resistance: 47 ohms ±10 percent, each coil.

Operate time:
10 milliseconds, maximum over temperature range (-001, -002, and -003) (includes contact bounce time).
6 milliseconds, maximum over temperature range (-004) (includes contact bounce time).

Coil power: 1 watt average, each coil.

Electrical data:

Insulation resistance: 1,000 megohms, minimum, except the resistance between coil and housing at high temperature shall be 500 megohms or greater.



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Dielectric withstanding voltage:

At atmospheric pressure: 60 Hz.

Between all mated contacts in the open position: 9 kV peak.

Between high voltage terminals and housing: 9 kV peak.

Between coil and housing: 500 V rms, 60 Hz.

Ground isolated: Ground isolated relays have operating elements insulated from the ground plane to provide absolute voltage isolation between the frame and the high voltage contacts.

Environmental data:

Temperature range: -55°C to +125°C.

Vibration: MIL-STD-202-204, test condition C (10-55 Hz, .06 inch DA/55-2000 Hz, 10 g).

Shock: MIL-STD-202-213, test condition J, 30 g, 11 milliseconds, half sine.

Physical:

Terminal strength: 5 pounds pull.

Dimensions and configuration: See figure 1.

Termination: Solder terminal.

Weight: 1 ounce, maximum.

Life test requirements:

Mechanical cycling: 1,000,000 cycles. Two samples units (cycling rate, 36,000 per hour maximum; contact current shall not exceed 10 milliamperes).

Quality assurance:

Dielectric withstanding voltage:

Tests to be conducted at atmospheric pressure rating only.

Duration of application: 5-10 seconds at a 10 percent increase in the dielectric withstanding voltage.

Part number: M83725/9 and a dash number from table I.

TABLE I. Part number and characteristics.

| Military part number | Mount |
|---|---|
| M83725/9-001 | Sleeve |
| M83725/9-002 | Flange |
| M83725/9-003 | Threaded base |
| M83725/9-004 | Sleeve |



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Referenced documents. In addition to MIL-DTL-83725, this document references the following:

MIL-STD-202-204 MIL-STD-202-213

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy – AS
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

Review activity:
Navy - OS

(Project 5945-2018-040)

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